



THE UNIVERSITY OF HARIPUR

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WEEK 4 INTERNSHIP PROGRESS REPORT

Remote Internship at FlyRank AI

STUDENT INFORMATION

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Department	Department of Information Technology
Program	BS Artificial Intelligence
Semester	7th Semester
Company	FlyRank AI
Internship Type	Remote Internship
Internship Duration	July 2026 – August 2026 (8 Weeks)
Report Period	Week 4 (July 22 – July 28, 2026)

Submitted To:

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1. Introduction

FlyRank AI is an AI-powered organic growth platform that helps businesses improve their visibility across traditional search engines and AI answer engines such as ChatGPT, Perplexity, and Gemini. The platform automates large parts of the content lifecycle - research, scoring, and optimization - primarily serving D2C, SaaS, and agency clients who need scalable organic growth.

This report covers Week 4 of my remote Machine Learning Engineering internship at FlyRank AI, under the Applied Search Intelligence track. It summarizes the assignment completed, the reasoning behind it, and the skills developed.

2. Objective of Week 4

Objective: To write a verified data contract for the chosen lane against FlyRank's real production-scale data warehouse, and to build the first transparent, rule-based baseline that any future model must be compared against.

3. Week 4: Data Contract & Baseline Scoring

Task Requirements: Write the data contract in plain words (unit of analysis, tables used, time window, target/proxy, and deliberate exclusions); verify every contract claim with real queries; build a small five-feature frame with justifications; demonstrate and then remove a deliberate data-leakage trap. Separately, check two real signals behind FlyRank's own flagging logic, encode one transparent scoring rule with a reason code and action label, generate a ranked queue, and manually review the top 10 results.

Research Performed: Connected directly to FlyRank's Hugging Face-hosted data warehouse (~79 million rows) using DuckDB, working on a mid-panel month (March 2026) rather than the sealed final test month, per the program's data-handling guidance. Studied the warehouse's table grain, join keys, and known data-quality caveats (such as unbalanced per-client history) before writing any queries.

Summary of Work Completed: Verified the data contract with three real queries: a grain check (confirmed one row per client-content-day, with zero duplicates), a row-count and date-span check (9.84 million rows across 55 clients and 331,437 content items for the month), and an availability check using IS TRUE filtering. Built a five-feature frame with a documented 'available at decision time' justification for each feature, and ran a deliberate leakage experiment - showing a model's score jump toward a near-perfect result when a label-derived column was added, then removing it and keeping the honest number. Separately, verified two real signals (content staleness and position-vs-CTR) against FlyRank's actual flagging logic, encoded one transparent baseline rule (stale-but-visible pages) with a reason code and action label, generated a ranked action queue directly from the notebook, and manually reviewed the top 10 flagged pages, noting for each what would make the flag wrong.

What Was Learned: Learned to verify every data assumption with an executable query rather than trust documentation alone, and experienced first-hand why a frozen, transparent baseline - not a model - must be the fixed point that all future modeling work is honestly measured against.

4. Skills Developed

Week 4 contributed to the development of the following technical, analytical, and professional skills:

Skill	Description
Data Contracts	Wrote and verified a formal data contract - unit of analysis, field classification, and time windows - using real queries.
SQL / DuckDB	Queried a ~79-million-row cloud data warehouse directly (Hugging Face + DuckDB) without downloading the full dataset.
Baseline Rule Design	Built a transparent, explainable scoring rule with reason codes as the honest benchmark for future models.
Leakage Awareness	Demonstrated, hands-on, how label-derived features artificially inflate model scores, and how to detect and remove them.

5. Challenges Faced

- › Real warehouse data access: Connecting to a ~79-million-row cloud warehouse required correctly configuring authentication tokens and adapting queries to the warehouse's actual column names, which differed from initial assumptions and had to be verified directly against the schema.
- › Avoiding data leakage: Ensuring that label-derived information never entered the feature set required deliberate testing (the leakage trap exercise) rather than relying on assumption alone.

6. Knowledge and Experience Gained

Week 4 provided hands-on experience working with production-scale data infrastructure, and reinforced why a verified data contract and a transparent, frozen baseline must both be in place before any model-building begins.

7. Conclusion

Week 4 completed the data-foundations phase of the internship: a verified data contract and a transparent, rule-based baseline built directly on FlyRank's real data warehouse. This baseline is now the honest benchmark that the model-building phase in the coming weeks must outperform.

Looking ahead, the objective for the next reporting period is to train and evaluate a machine learning model against this baseline using a proper client-holdout validation split.